Engineering Change Notice(ECN)

ECN-D-2604-04(Application of ESD to Single Robot Pusher Tip)

FA연구소 설계2팀

FA Research ME 2 team

Written	Review	Confirm
김문선	최동수	이태경
04/03	04/03	04/03

NINEBELL

1. Engineering Change Request

NINEBELL

설계 변경 요청서(ECR)
(Engineering Change Request)

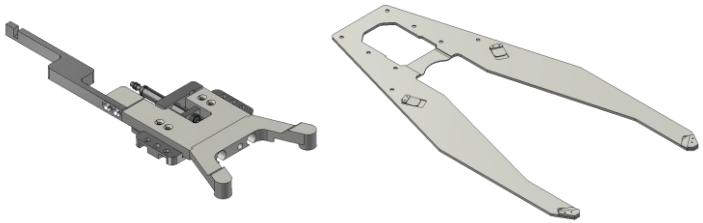
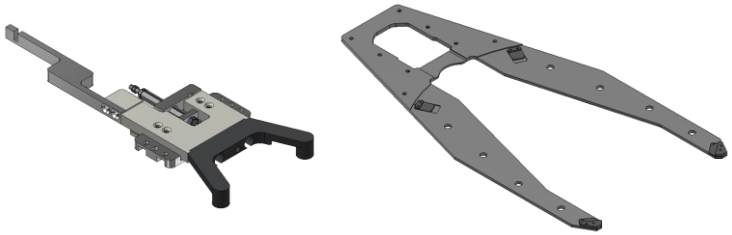
				Customer	ACM
				ECR No.	
Requesting department	Name / Department	/	Change ITEM	☐ Customer Requst ☐ Cost reduction ☐ Quality improve ☐ Work Efficiency ☐ Parts discontinued ☐ ETC	
	Date	2025-07-21			
Request project	Category	Project Type		변경 Level (Change Level)	☑ Level 1 Design structure changes are required
	☐ MP ☐ ES				☐ Level 2 Simple part and work method change

Before(Problems and Change Requests)	After(Expected Effect)
原有pusher tip不具备ESD The original push tip does not have ESD.	增加ESD功能 Please add ESD function to Pusher TIP.

Related Division		Potential Issue	
	☑ Mechanical design		☐ Instrument & tool additional cost incurred
	☐ Electrical design		☐ Work process change and work time
	☐ Manufacturing		☐ Mechanical failure
	☐ Purchasing & Materials		☐ ETC
	☐ Quality		

2. Engineering Change Notice

<table>
<tr><td>적용 장비 타입
Applicable project type</td><td colspan="2">SINGLE ROBOT</td><td>적용 범위
Scope</td><td>■ 신규 발주분 (New order)
□ 기발주분 (Previous order)
□ 기출하장비 (Existing shipped equipment)</td></tr>
<tr><td rowspan="2">변경 사유
Reason for change</td><td colspan="2" rowspan="2">■ 고객 요청 (Customer request) □ 품질개선 (Quality improve)
□ 원가절감 (Cost reduction) □ 생산성 향상 (Increase productivity)
□ 부품 단종 (Part discontinuation) □ 기타 (ETC)</td><td>ECR NO.</td><td>ECR-S-2604-04</td></tr>
<tr><td>ECN NO.</td><td>ECN-D-2604-04</td></tr>
<tr><td>변경 상세 사유
Detailed reason for change</td><td colspan="4">Wafer 와 닿는 모든 파트는 ESD를 적용 할 것.
위 내용에 근거하여 SINGLE ROBOT은 별도에 사양 협의가 없더라도 HAND, HAND TIP, PUSHER TIP 에 모두 ESD(10^6 ~ 10^9 Ω)를 적용함.

To ensure wafer safety, ESD protection is required for all parts in direct contact with the wafer. Accordingly, ESD specification(10^6 ~ 10^9 Ω) are being applied to the HAND, HAND TIP, PUSHER TIP of the Single Robot as a standard, even in the absence of separate specification agreements</td></tr>
<tr><td rowspan="2">변경 사항
Changes</td><td colspan="2">변경 전(Before)</td><td colspan="2">변경 후(After)</td></tr>
<tr><td colspan="2">  </td><td colspan="2">  </td></tr>
<tr><td>상세내용
Detail</td><td colspan="2">HAND : CERAMIC
HAND TIP : PEEK
PUSHER TIP : PEEK</td><td colspan="2">HAND : CERAMIC (ESD 10^6—10^9 Ω)
HAND TIP : ESD PEEK
PUSHER TIP : ESD PEEK</td></tr>
</table>

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2. Engineering Change Notice

전체 이미지 Full Image	단품 로봇 특성상 핸드, 핸드 팁, 푸셔 팁 종류가 다양하여 “기존 품번 -> 변경 품번” 내용 공유 자체가 무의미하여 품번 기재를 생략. 이 ECN을 근거로 JC03-098 이후 PO장비는 모두 ESD를 적용. Due to the diverse variety of HAND, HAND TIP, PUSHER TIPs for Single Robot models, providing a direct comparison of “Old P/N → New P/N is redundant; therefore, specific Part Numbers are omitted from this document. Based on this ECN, all equipment ordered under Pos following JC03-098 will have ESD protection applied as a standard. ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	Y / (N)
승인 날짜 Approval date	1. Ninebell : 2026.04.07 2. Customer : / /		나인벨 승인 Ninebell approval	QC 2026.04.07 PASS	고객사 승인 Customer approval	ACM confirm 4/7/26
적용 일자 Effective date	JC03-098 이후 PO 장비 적용 Effective for all Pos following JC03-098		최초 적용 장비 Initially Applied Equipment	HP87-007 CW49-093 CW49-094 JC03-098 ECN 작성일 기준 현재 PO나온 장비에 모두 적용 Applied to all current PO equipment including : HP87-007 CW49-093 CW49-094 JC03-098 (As of the ECN preparation date)		

3. Attachments

N/A